EXP NO:17 EXPERIMENT TO UNDERSTAND LINEAR REGRESSION FOR A GIVEN DATA SET

CODE :

```
import numpy as np
import pandas as pd
import pickle
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
df = pd.read_csv('Salary_data.csv')
print("Before removing missing values:")
df.info()
df.dropna(inplace=True)
print("\nAfter removing missing values:")
df.info()
print(df.describe())
features = df.iloc[:, [0]].values
label = df.iloc[:, [1]].values
x_train, x_test, y_train, y_test = train_test_split(
    features,
    label,
    test_size=0.2,
    random_state=42
)
```

```
model = LinearRegression()
model.fit(x_train, y_train)
print("\nTraining R² score:", model.score(x_train, y_train))
print("Testing R² score:", model.score(x_test, y_test))
print("Coefficient:", model.coef_)
print("Intercept:", model.intercept_)
with open('/content/SalaryPred.model', 'wb') as file:
    pickle.dump(model, file)
with open('/content/SalaryPred.model', 'rb') as file:
    model = pickle.load(file)
yr_of_exp = float(input("Enter Years of Experience: "))
yr_of_exp_np = np.array([[yr_of_exp]])
salary = model.predict(yr_of_exp_np)
print(
    f"Estimated salary for {yr_of_exp} years of experience "
    f"is ₹{salary[0][0]:,.2f}"
)
```

OUTPUT :

```
Before removing missing values:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 300 entries, 0 to 299
Data columns (total 2 columns):
```

```
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   YearsExperience  296 non-null    float64
 1   Salary          296 non-null    float64
dtypes: float64(2)
memory usage: 4.8 KB

After removing missing values:
<class 'pandas.core.frame.DataFrame'>
Index: 292 entries, 0 to 299
Data columns (total 2 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   YearsExperience  292 non-null    float64
 1   Salary          292 non-null    float64
dtypes: float64(2)
memory usage: 6.8 KB
       YearsExperience         Salary
count       292.000000     292.000000
mean          7.839384   65975.287671
std           4.076998   22387.172902
min           0.300000   18000.000000
25%           4.575000   49128.500000
```

50% 7.900000 65040.000000

75% 11.200000 82683.500000

max 15.000000 121956.000000

Training R^2 score: 0.907767542415078

Testing R^2 score: 0.9098570405486877

Coefficient: [[5245.51746903]]

Intercept: [24614.15597493]

Enter Years of Experience: 3

Estimated salary for 3.0 years of experience is ₹40,350.71